Virtual Machines and Runtime Systems – Organizational Matters

Armin Größlinger

Winter semester 2026/2027

Lecturer

Lecture & Tutorials:	Armin Größlinger
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Dates

Lecture:	Tuesday 16.00–17.30 in (IM) SR 007 – start at 16.00 OK?
Tutorial:	Wednesday 12.15–13.45 in (IM) HS 12
Examination:	written exam 1st exam: Thursday, Feb 11, 2027 at 14.00–15.30 2nd exam: Thursday, Apr 01, 2027 at 14.00–15.30 (rooms to be announced in Stud.IP) duration: 90 minutes Do not forget to register in time via Campus Portal!

See `https://www.uni-passau.de/en/examinations-office/semester-related-information` for registration periods, etc.

The lecture can be used with 6 ECTS points as module in

- Master Informatik (Modulgruppe "‘Programmierung und Softwaresysteme"’) / M.Sc. Computer Science (Focus "‘Programming and Software Systems"’).

Contents of the Lecture

- Introduction to virtualization
- Runtime systems
  - C runtime system
  - processes
  - thread scheduling
  - memory management
  - runtime systems for high-level languages
- Virtual machines (VMs)
  - types: system VMs, process VMs, high-level language VMs
  - implementation (interpreter, compiler, garbage collection, . . .)
  - instrumentation
  - applications
- Containers (“light-weight virtualization”)
  - Linux namespaces
  - Kubernetes

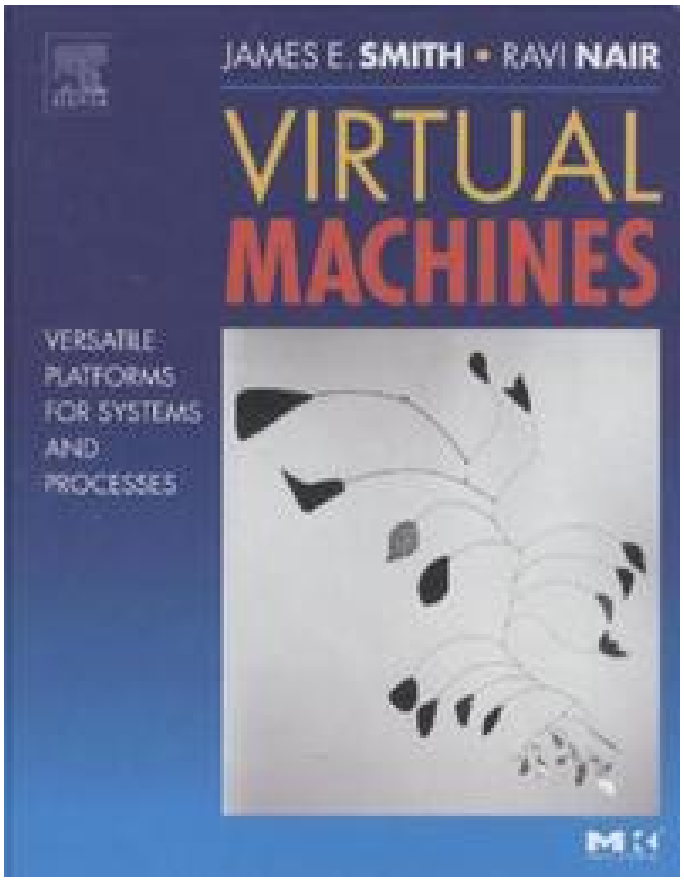
Some Notes about the Tutorials

In the tutorials, we will use C and C++, maybe Rust.

Who is (not) familiar with C / C++ / Rust ?

In the tutorial session on October 21, there will be a quick introduction to (some aspects of) C/C++. (There is no tutorial session on October 14.)

There is an online book for the C language: *The C Book*,
`http://publications.gbdirect.co.uk/c_book/`
For C++, there is `http://www.cplusplus.com/`

Literature

J. Smith, R. Nair, Virtual Machines, Morgan Kaufmann, 2005